CNENGR115 Computers and Control

First Order Systems (2)

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Control Lectures

Part 1 – Introduction

Part 2 – Mathematical Models

Part 3 – Control Actions

Standard Notation for a 1st Order System

$$\tau \dot{x} + x = Ku(t) \quad \text{OR} \quad \boldsymbol{\tau} \frac{dx}{dt} + x = Ku(t)$$

output variable x

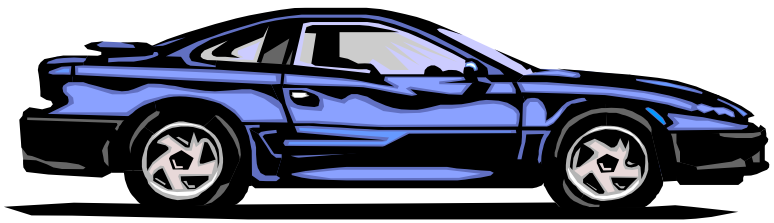
input variable $u(t)$

time constant τ

steady state gain K

Speed – Acceleration

$$v \longrightarrow \dot{v}$$



Analysis of Vehicle Speed Model

- This is our model for vehicle speed

$$m\dot{v} + bv = u(t)$$

- By convention, the model is often re-arranged as follows

$$\frac{m}{b}\dot{v} + v = \frac{1}{b}u(t)$$

- Compare with the general 1st order model

$$\tau\dot{x} + x = Ku(t)$$

Output

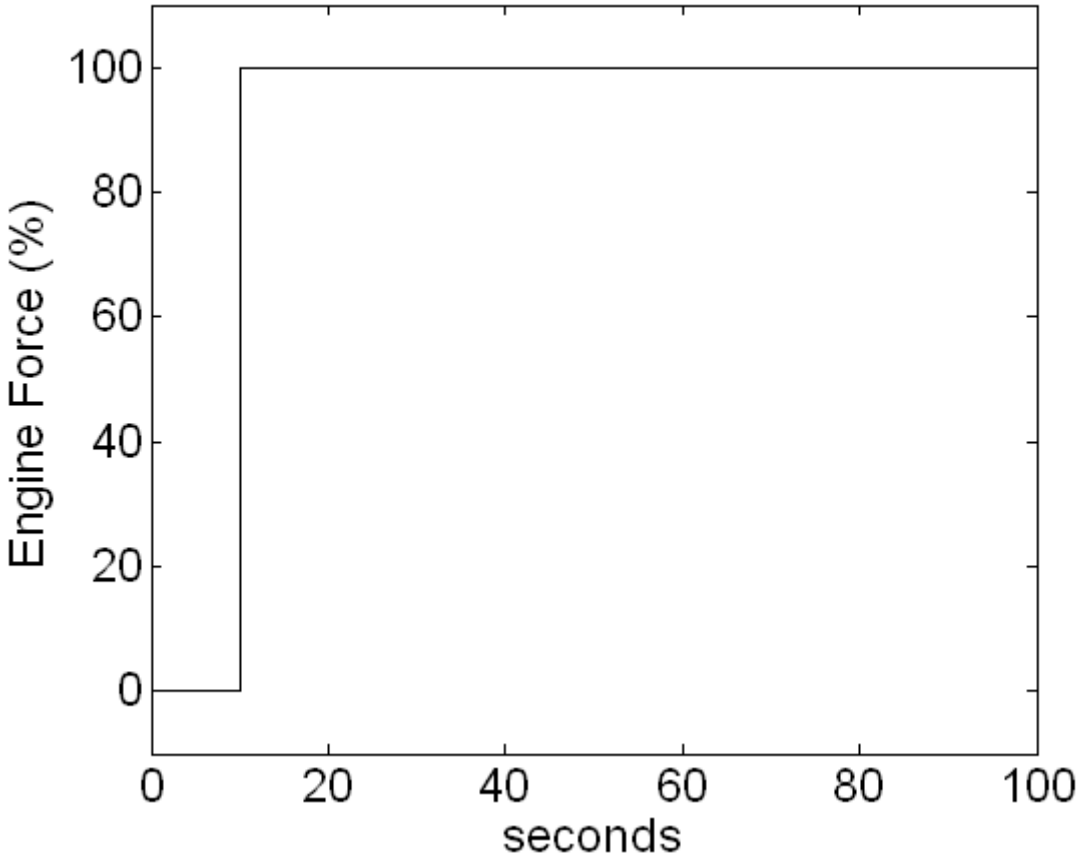
Input

Time constant

Steady state gain

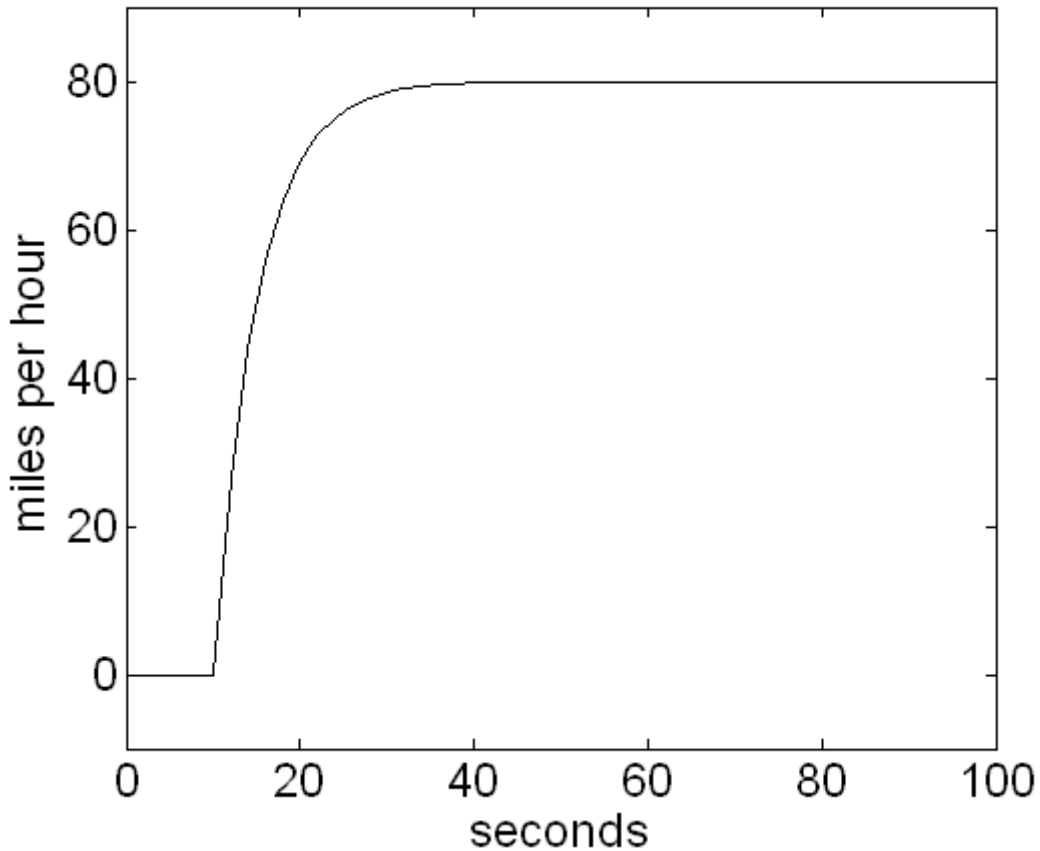
Engine Force Step Input

- In this experiment, the engine force is a **step input** from zero to 100%
- Assume that 0% is zero force and that 100% represents full power



Vehicle Speed Step Response

- Given typical values for mass and friction, the response of the speed to a 100% step in engine power is shown right
- A **simulation package** was used to solve the differential equation and obtain this graph (Matlab/Simulink)



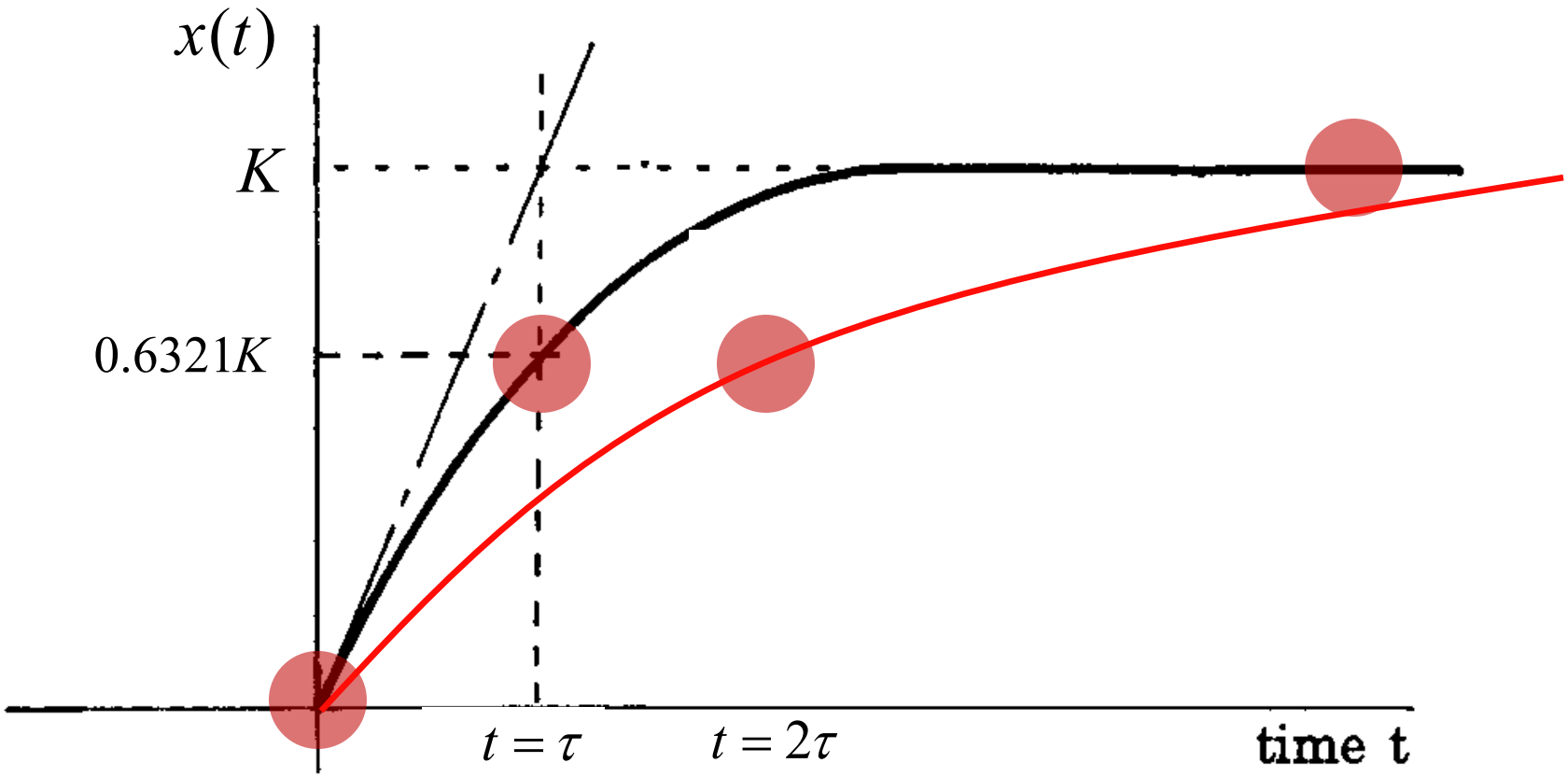
Unit Step Response of a 1st Order System

- If the input is increased from zero to 1, this is called a unit **step input**
- In this case, it can be shown that the **output** behaves as shown right

$$x(t) = K\left(1 - e^{-t/\tau}\right) \mathrm{u(t)}$$

Time t	Output $x(t)$
$t = 0$	$x(0) = K(1 - e^{0}) = 0$
$t = \tau$	$x(\tau) = K(1 - e^{-1}) = 0.6321K$
$t \to \infty$	$x(\infty) = K(1 - e^{-\infty}) = K$

Unit Step Response of a 1st Order System



Steady State Gain

- For a unit step input, the output gradually approaches the **steady state gain** of the system

$$x(t \to \infty) = K$$

- More generally, for a constant input

$$x(t \to \infty) = K \cdot u(t \to \infty)$$

- To find the steady state gain from the step response graph

$$K = \frac{x(t \to \infty)}{u(t \to \infty)}$$

Time Constant

- The **time constant** of the system is a measure of the speed of response
- For a unit step input

$$x(t=\tau)=0.63K=0.63x(t\to\infty)$$

- If the input is non-unity

$$x(t=\tau)=0.63K\cdot u\left(t\to\infty\right)=0.63x(t\to\infty)$$

- To find the time constant from the step response graph, measure the time it takes for the output to reach 63% of its final value

1st Order Model for Vehicle Speed

- Earlier we developed a 1st order **mechanistic** model for vehicle speed with two parameters as follows

$$\tau = \frac{m}{b} \quad \text{and} \quad K = \frac{1}{b}$$

- The model coefficients may alternatively be estimated directly from data

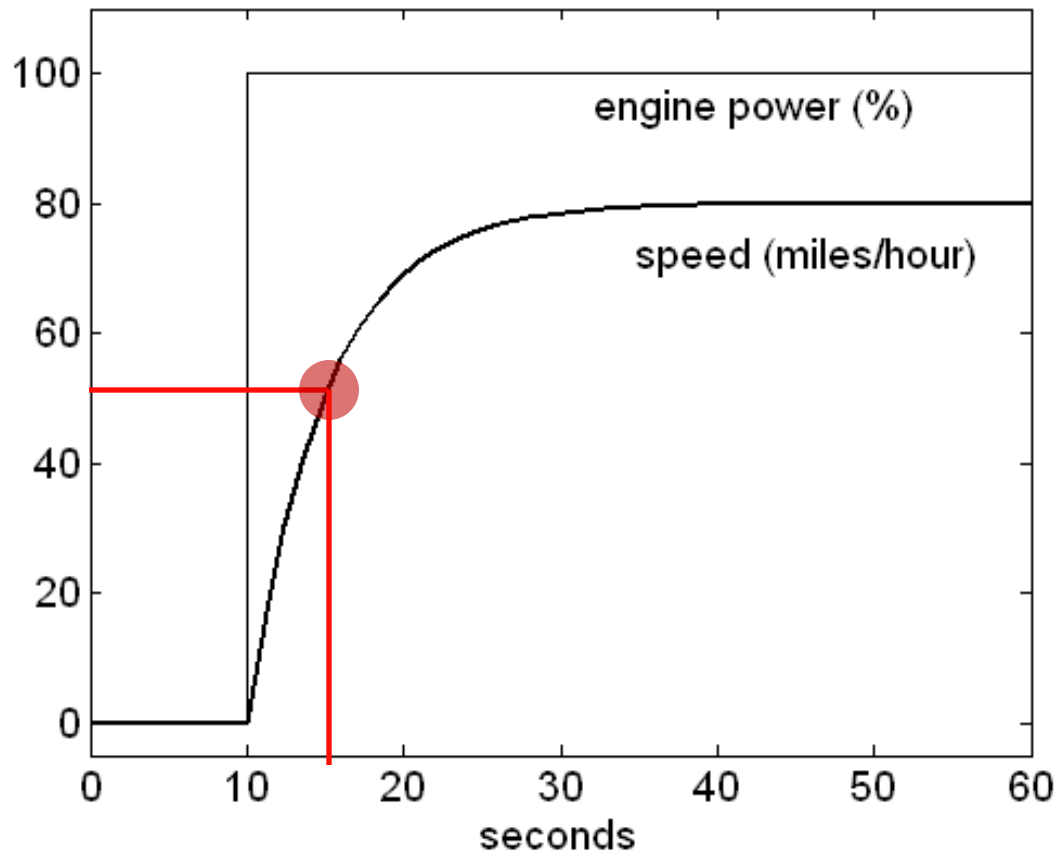
1st Order Model for Vehicle Speed

$$\tau\dot{x} + x = Ku(t)$$

$$5\dot{x} + x = 0.8u(t)$$

$$K = \frac{x(t \to \infty)}{u(t \to \infty)}$$

$$K = \frac{80}{100} = 0.8$$



$$x(t = \tau) = 0.63Ku\left(t \to \infty\right) = 0.63 \times 80 = 50.4\,\text{mph} \qquad \tau = 5s$$

1st Order Model for Ventilation Rate

- Analysis of experimental data suggests that the ventilation rate produced by an axial fan can be represented by $\tau\dot{x} + x = Ku(t)$
- This is the same model **structure** as the that developed for vehicle speed!
- The coefficients can be estimated directly from experimental data using this graphical approach, i.e. a **black box** approach
- Advanced subject – statistical methods (e.g. least squares) can be used to estimate the coefficients of more complex models

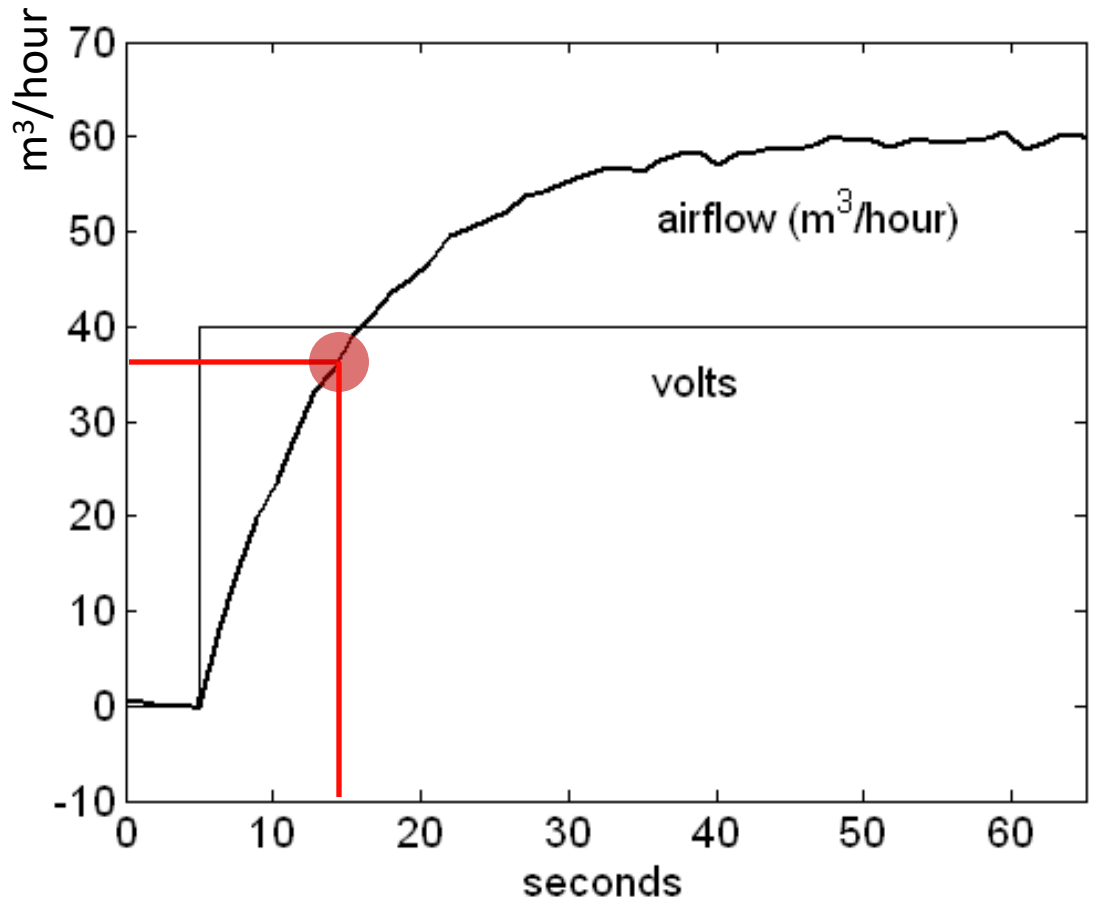
1st Order Model for Ventilation Rate

$$\tau\dot{x} + x = Ku(t)$$

$$10\dot{x} + x = 1.5u(t)$$

$$K = \frac{x(t \to \infty)}{u(t \to \infty)}$$

$$K = \frac{60}{40} = 1.5$$



$$x(t=\tau) = 0.63Ku\left(t \to \infty\right) = 0.63 \times 60 = 37.8 \text{ m}^3 / \text{hour} \qquad \tau = 10s$$

1st Order Model for Ventilation Rate

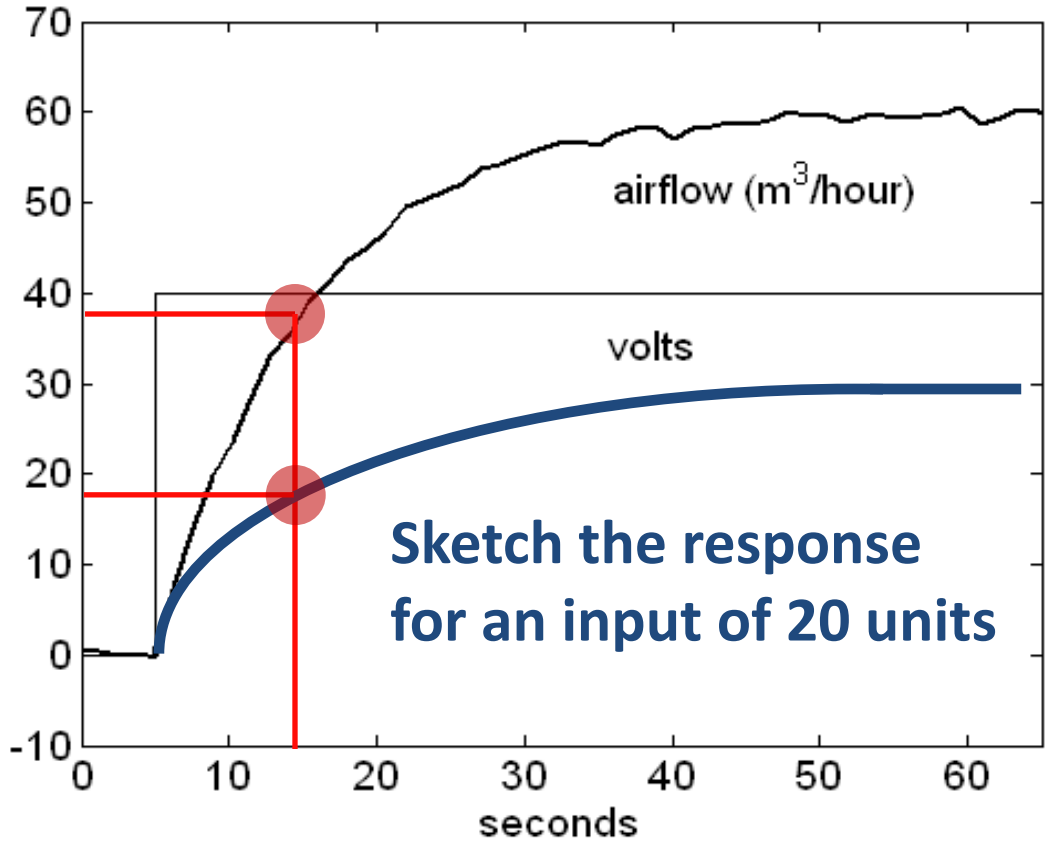
$$\tau\dot{x} + x = Ku(t)$$

$$10\dot{x} + x = 1.5u(t)$$

$$K = \frac{x(t \to \infty)}{u(t \to \infty)}$$

$$K = \frac{60}{40} = 1.5$$

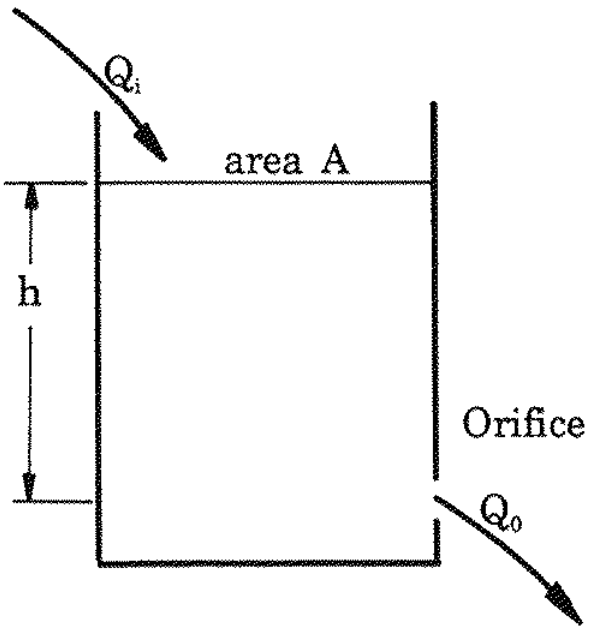
$$x(t \to \infty) = 1.5 \times u$$



$$x(t = \tau) = 0.63Ku\left(t \to \infty\right) = 0.63 \times 30 = 18.9 \text{ m}^3 / \text{hour} \qquad \tau = 10s$$

1st Order Model for Hydraulic System

- **Single Tank Hydraulic System**
- Assumptions
  - Flow rate proportional to head
  - No energy dissipated in orifice
  - Incompressible fluid



1st Order Model for Hydraulic System

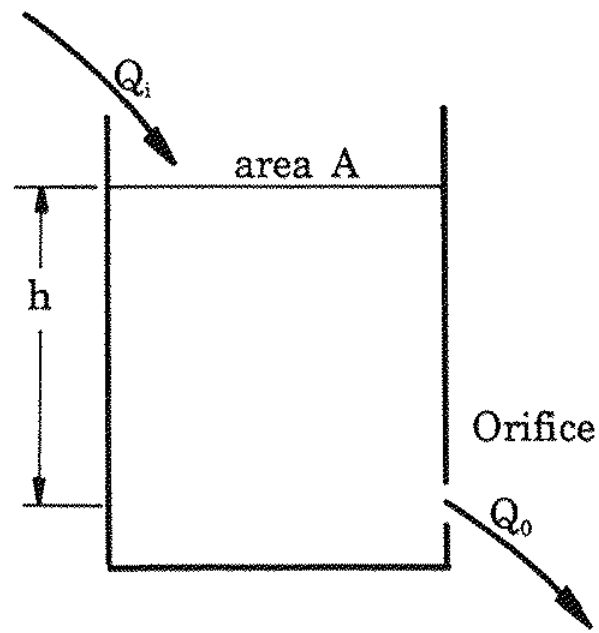
- From the **1st assumption**

$$Q_o = Kh \rightarrow h = \frac{Q_o}{K} \rightarrow \frac{dh}{dt} = \frac{1}{K}\frac{dQ_o}{dt}$$

- Using **Torricelli's law**

$$\frac{dV}{dt} = A\frac{dh}{dt} = Q_i - Q_o$$

$$\frac{A}{K}\frac{dQ_o}{dt} = Q_i - Q_o \rightarrow \frac{A}{K}\frac{dQ_o}{dt} + Q_o = Q_i$$



1st Order Model for Hydraulic System

$$\frac{A}{K}\frac{dQ_o}{dt} + Q_o = Q_i$$

Output

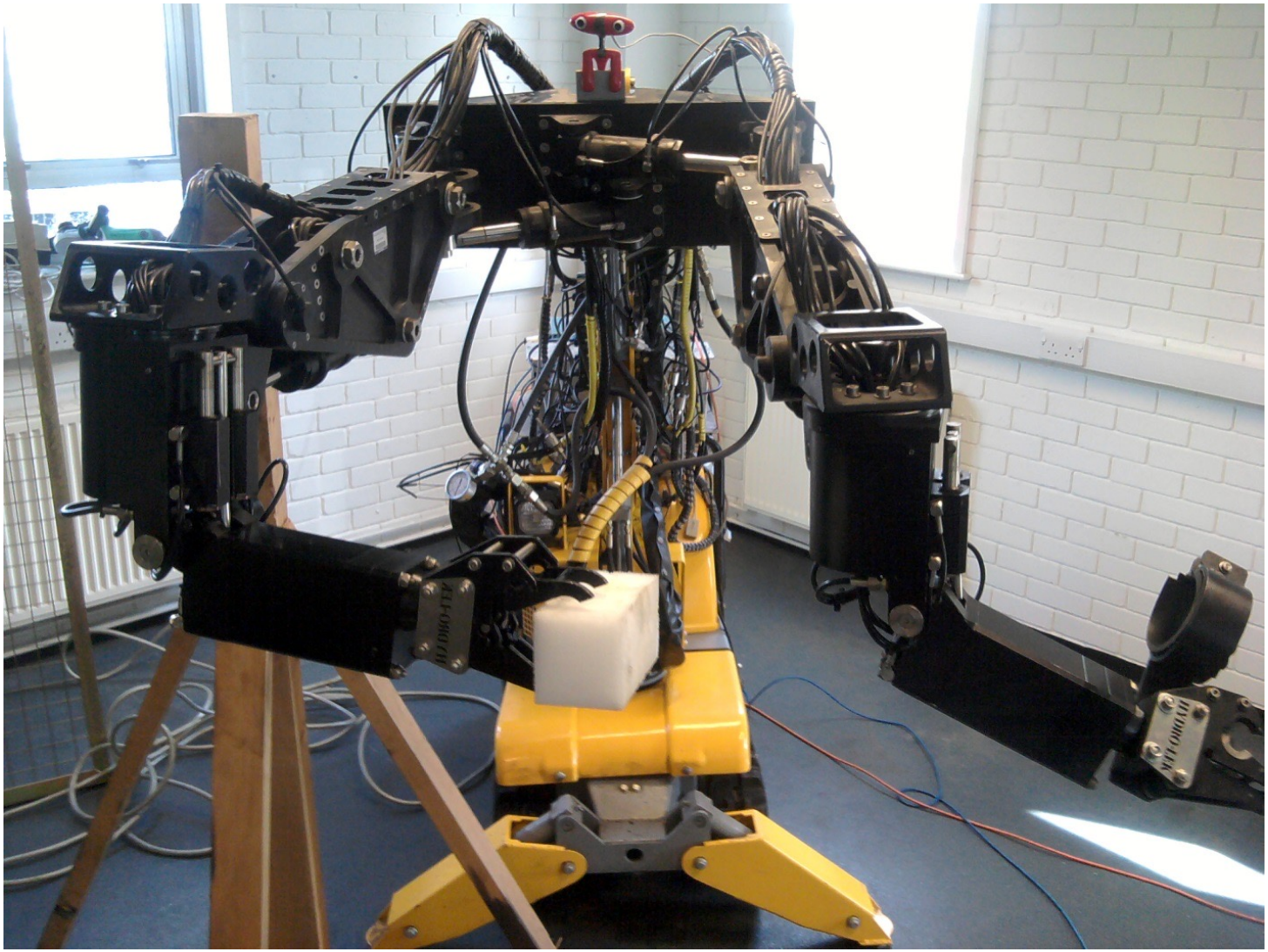
Input

Time constant

Steady state gain=1

$$\tau \frac{dx}{dt} + x = Ku(t)$$

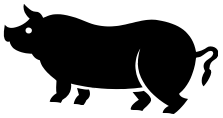
1st Order Model for Hydraulic System



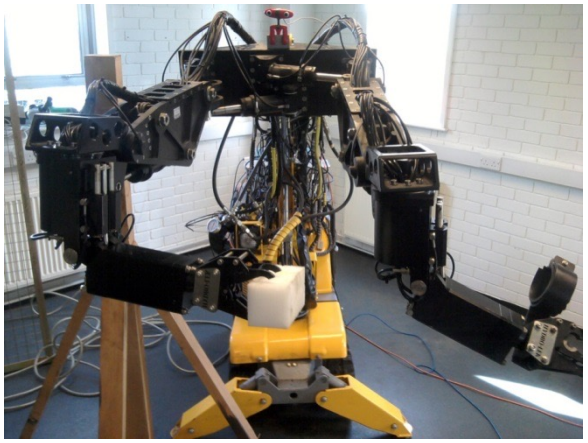
Why use the Generalised Form?

$$\tau\dot{x} + x = Ku(t)$$

- We have developed a model for vehicle speed and converted it into the general 1st order form
- Ventilation rate can be described by the same model structure



- Many other engineering systems can also be represented using this model
  - **Mechanical** systems
  - **Thermal** systems
  - **Electrical** systems
  - **Hydraulic** systems



- One advantage of the general form is that the same analysis applies to all 1st order systems, e.g. the **step response** can be used to estimate the parameters of these systems

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